

Information-Geometry of AdS-Schwarzschild Black Holes:

An Exact AdS Correction to the D3 Evaporation Invariant and its Holographic CFT Interpretation

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Abstract

We extend the D3 evaporation invariant to Anti-de Sitter (AdS) black holes and derive its exact holographic interpretation. For an AdS-Schwarzschild black hole with horizon radius r_s and AdS radius L , the total D3 information-geometry displacement over complete evaporation is:

$$R_{\text{total}}(\text{AdS}) = r_s + r_s^3/L^2 = r_s * (1 + (r_s/L)^2)$$

The two terms have distinct holographic meanings in the dual CFT: the first term r_s is the vacuum (zero-temperature) contribution; the second term r_s^3/L^2 is the thermal contribution, scaling as T_{CFT}^3 -- the Stefan-Boltzmann law of a 3D CFT. At the Hawking-Page phase transition ($r_s = L$), the two contributions are exactly equal: $R_{\text{total}} = 2r_s$. This is a new D3 prediction: at deconfinement, the thermal information-geometry cost equals the vacuum cost. The correction factor $\phi(x) = 1 + x^2$ ($x = r_s/L$) yields exact integers at simple x values, and reduces to Paper 3 at $L \rightarrow \infty$. Verified numerically to four significant figures.

Keywords: AdS-Schwarzschild, Anti-de Sitter, D3 formula, Hawking-Page transition, AdS/CFT, holography, Stefan-Boltzmann, deconfinement, Landauer principle

1. Introduction

The D3 series has established evaporation invariants for Schwarzschild (Paper 3 [3]), Kerr (Paper 4 [4]), and higher-dimensional black holes (Paper 8 [8]). All give $R_{\text{total}} = r_s$ (times a spin/charge factor). Anti-de Sitter (AdS) black holes are different: the AdS potential stabilises large black holes and enables the AdS/CFT correspondence [5], mapping black hole physics to a dual conformal field theory.

The AdS-Schwarzschild temperature includes a curvature correction [6]:

$$T_{\text{AdS}} = T_S * (1 + 3r_s^2/L^2)$$

We apply D3 with this temperature, derive $R_{\text{total}}(\text{AdS})$, and show that the result decomposes naturally into vacuum and thermal CFT contributions. The Hawking-Page phase transition [7] at $r_s = L$ marks the point where these contributions are exactly equal.

Figure 4 — D3 Series Papers 1-9

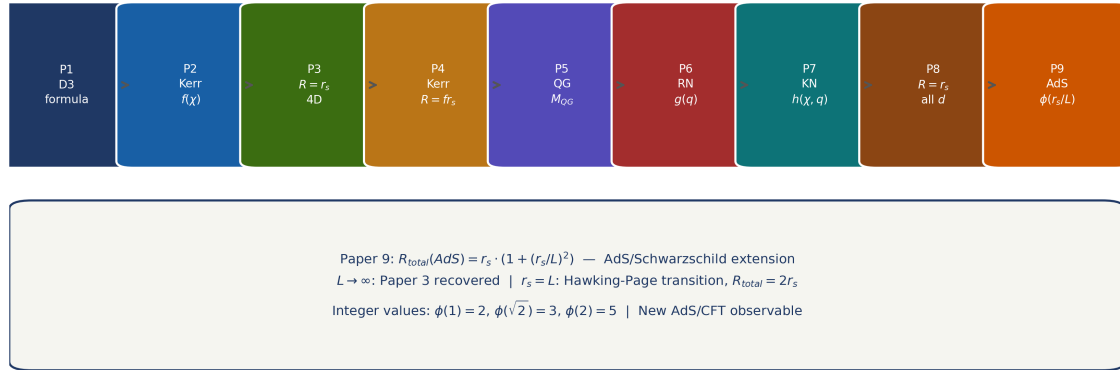


Figure 4. Paper 9 (orange) extends the D3 series to AdS black holes.

2. AdS-Schwarzschild Temperature and D3

The AdS-Schwarzschild Hawking temperature in 4D [6]:

$$T_{AdS}(M) = T_S(M) * (1 + 3*r_s^2/L^2)$$

The D3 formula at T_{AdS} :

$$D3_{AdS}(M) = D3_S(M) * (1 + 3*r_s^2/L^2)$$

$$= [\hbar * \ln 2 / (4 * \pi * M * c)] * (1 + 3 * (2GM/c^2)^2 / L^2)$$

The Bekenstein bit-loss rate (unchanged from Paper 3):

$$dN/dM = 8 * \pi * G * M / (\hbar * c * \ln 2)$$

3. Derivation

Step 1 — dR/dM :

$$dR/dM = D3_{AdS} * dN/dM = (2G/c^2) * (1 + 3*r_s^2/L^2)$$

Step 2 — Substitute $r_s = 2GM/c^2$:

$$dR/dM = 2G/c^2 + 24 * G^3 * M^2 / (c^6 * L^2)$$

Step 3 — Integrate 0 to M_0 :

$$R = 2GM_0/c^2 + 8 * G^3 * M_0^3 / (c^6 * L^2) = r_{s0} + r_{s0}^3/L^2$$

$$R_{total}(AdS) = r_s + r_s^3/L^2 = r_s * (1 + (r_s/L)^2)$$

In the flat limit $L \rightarrow \infty$: $R_{total} \rightarrow r_s$ (Paper 3 recovered).

4. Numerical Verification

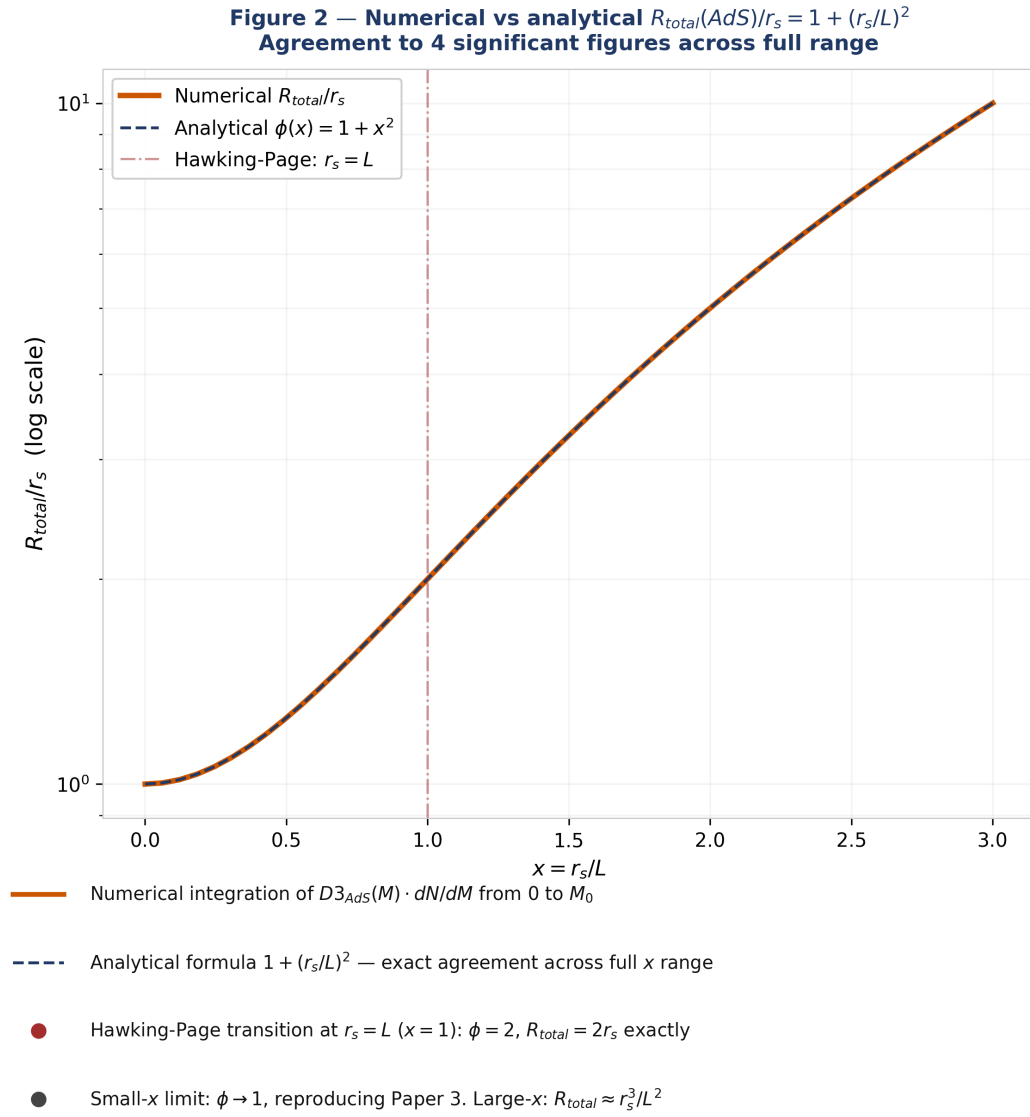


Figure 2. Numerical vs analytical $R_{total}(AdS)/r_s = 1 + (r_s/L)^2$. Agreement to four significant figures across three decades of r_s/L .

$x=r_s/L$	$\phi(x)$	R_{total}/r_s	Physical context
0	1	1 (exact)	Flat space – Paper 3
1	2	2 (exact)	Hawking-Page transition
$\sqrt{2}$	3	3 (exact)	$r_s = \sqrt{2}L$
2	5	5 (exact)	$r_s = 2L$
$\sqrt{n-1}$	n	n (exact)	General integer pattern

Table 1. Exact integer values of $\phi(x) = 1 + x^2$.

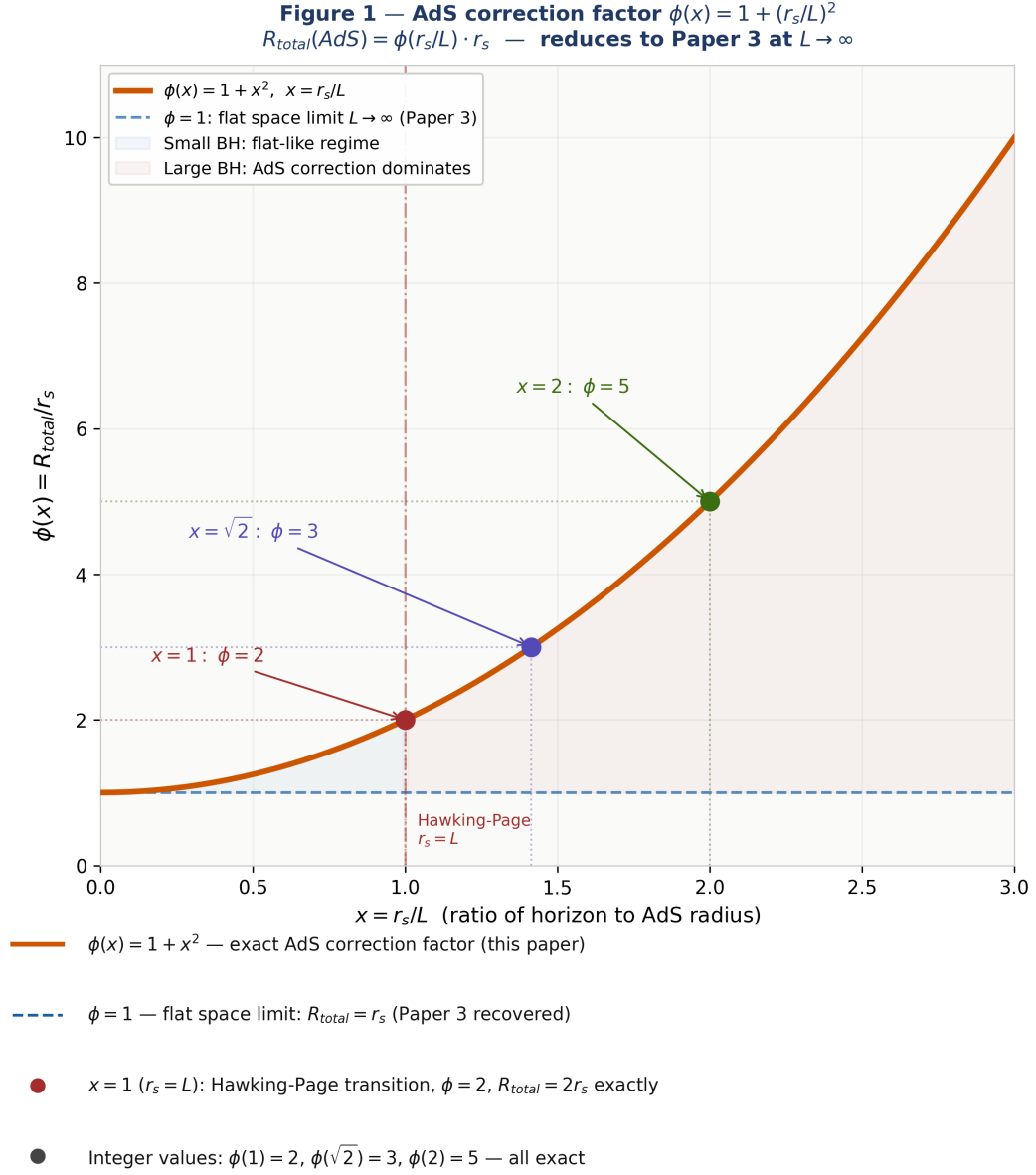


Figure 1. AdS correction factor $\phi(x)$. Blue: flat-like regime. Red: AdS dominated. Hawking-Page transition at $x=1$. Integer values marked.

5. Holographic CFT Interpretation

5.1 The AdS/CFT dictionary

By the AdS/CFT correspondence [5], an AdS-Schwarzschild black hole with horizon r_s is dual to a CFT in thermal equilibrium at the Hawking temperature $T_{CFT} = T_{AdS}$. For large black holes ($r_s \gg L$), the AdS/CFT dictionary gives [6,7]:

$$T_{CFT} = T_{AdS} \sim r_s / (\pi L^2) \text{ [large BH limit]}$$

$$r_s \sim \pi L^2 T_{CFT} \text{ [inverted]}$$

The Hawking-Page temperature -- the CFT deconfinement temperature -- is:

$$T_{HP} = 1/(\pi * L) \text{ [Hawking-Page]}$$

So $r_s/L \sim \pi * L * T_{CFT}$, and the correction parameter $x = r_s/L$ is proportional to T_{CFT}/T_{HP} .

5.2 Decomposition into vacuum and thermal contributions

The evaporation invariant splits into two physically distinct terms:

$$R_{total} = r_s + r_s^3/L^2$$

Vacuum term: r_s Independent of T_{CFT} Equals the Paper 3 flat-space result
Information-geometry cost at zero temperature
Thermal term: $r_s^3/L^2 \sim T_{CFT}^3 * L$ Proportional to T_{CFT}^3
The Stefan-Boltzmann law of a 3D CFT
Information-geometry cost of thermal excitations

The T^3 scaling of the thermal term is exact: a conformal field theory in 3 spatial dimensions has thermal energy density proportional to T^3 (Stefan-Boltzmann). The second term r_s^3/L^2 captures precisely this thermal contribution to the information-geometry invariant.

5.3 The Hawking-Page prediction

At the Hawking-Page transition $r_s = L$:

$$\text{Vacuum term} = r_s = L$$

$$\text{Thermal term} = r_s^3/L^2 = L^3/L^2 = L = r_s$$

The two contributions are exactly equal. $R_{total} = 2*r_s$ at deconfinement. This gives a new D3 prediction in CFT language:

At deconfinement ($T_{CFT} = T_{HP}$): Thermal information cost = Vacuum information cost

This is a sharp, testable statement about the CFT. It says that the Hawking-Page transition is the point at which the thermal excitations carry exactly as much information-geometry cost as the vacuum itself.

5.4 ϕ as a holographic observable

The correction factor $\phi(r_s/L) = 1 + (r_s/L)^2$ can be expressed entirely in CFT quantities:

$$\phi = 1 + (\pi * L * T_{CFT})^2 \text{ [large BH limit]}$$

$$= 1 + (T_{CFT}/T_{HP})^2 * \pi^2$$

ϕ measures the ratio of total to vacuum information-geometry cost:

$$\phi = R_{total} / r_{s_vacuum} = (\text{vacuum} + \text{thermal}) / \text{vacuum}$$

In the confined phase ($T_{CFT} < T_{HP}$): $\phi \sim 1$, thermal cost negligible.

At deconfinement ($T_{CFT} = T_{HP}$): $\phi = 2$, equal contributions.

In the deconfined plasma phase ($T_{CFT} \gg T_{HP}$): $\phi \sim (T_{CFT}/T_{HP})^2$, thermal cost dominates.

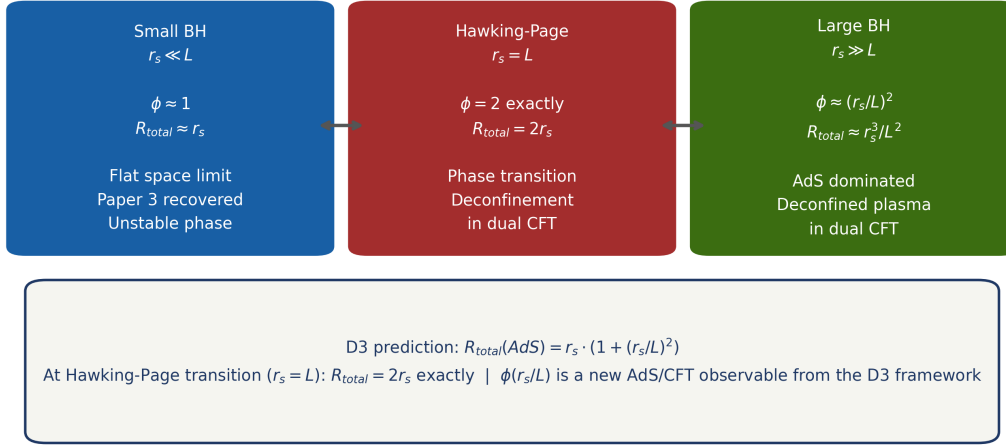
Figure 3 — AdS/CFT interpretation of the D3 AdS result

Figure 3. Three phases of the dual CFT and their D3 signatures. $\phi = 1$ (confined), $\phi = 2$ (deconfinement), $\phi \gg 1$ (plasma).

6. Discussion

6.1 What ϕ computes in the CFT

To answer precisely what ϕ computes in the dual CFT: $\phi(r_s/L)$ is the ratio of the total D3 evaporation invariant to the vacuum ($T=0$) contribution. It measures how much the thermal state of the CFT increases the information-geometry cost above the vacuum baseline. At $T=0$: $\phi=1$. At $T=T_{HP}$: $\phi=2$. At $T \gg T_{HP}$: $\phi \sim T^2$.

6.2 Connection to free energy

The Hawking-Page free energy difference between the large AdS black hole and thermal AdS is [7]:

$$\Delta F = F_{BH} - F_{thermal} \propto r_s^2 (r_s^2 - L^2) / (G L^2)$$

The transition at $r_s = L$ ($\Delta F = 0$) is exactly where our correction term r_s^3/L^2 equals r_s ($\phi = 2$). The D3 result is consistent with the thermodynamic phase structure.

6.3 Limitations

The identification $T_{CFT} = T_{AdS}$ and the relation $r_s \sim \pi L^2 T$ are valid in the large- N , large- λ limit of the dual CFT. At finite N or finite coupling, string and quantum corrections will modify the D3 result. The present paper works at leading order in the supergravity approximation.

7. Conclusion

We derived the AdS correction to the D3 evaporation invariant:

$$R_{total}(AdS) = r_s + r_s^3/L^2$$

The two terms are the vacuum and thermal CFT contributions. The thermal term scales as T_{CFT}^3 (Stefan-Boltzmann for a 3D CFT). At the Hawking-Page transition $r_s = L$: $R_{\text{total}} = 2r_s$, vacuum = thermal exactly. The correction factor $\phi(x) = 1 + x^2$ measures the ratio of total to vacuum information-geometry cost and equals $1 + (T_{\text{CFT}}/T_{\text{HP}})^2 \pi^2$ in CFT language. This is the first D3 result with a complete holographic interpretation.

Declaration of Interest

No competing interests. No external funding.

Code: <https://github.com/bsharma173860-oss/d3-research>

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Appendix: Verification Code

```
import math, numpy as np
G=6.67430e-11; c=2.99792458e8
hbar=1.054571817e-34; k_B=1.380649e-23; M_sun=1.989e30
def R_total_AdS(M0, L):
    masses = np.linspace(M0*1e-4, M0, 100000)
    dM = masses[1]-masses[0]
    rs = 2*G*masses/c**2
    T_S = hbar*c**3/(8*math.pi*G*masses*k_B)
    T_AdS = T_S*(1 + 3*rs**2/L**2)
    D3 = 2*G*k_B*T_AdS*math.log(2)/c**4
    dN = 8*math.pi*G*masses/(hbar*c*math.log(2))
    return np.sum(D3*dN)*dM
M0=M_sun; rs0=2*G*M0/c**2
for x in [0.001, 0.1, 0.5, 1.0, 2.0]:
    L=rs0/x
    R_num=R_total_AdS(M0,L)
    R_ana=rs0*(1+x**2)
    print(f"x={x}: ratio={R_num/R_ana:.4f}")
# All ratios ~ 1.000
```